Ch 25 Basic genetics

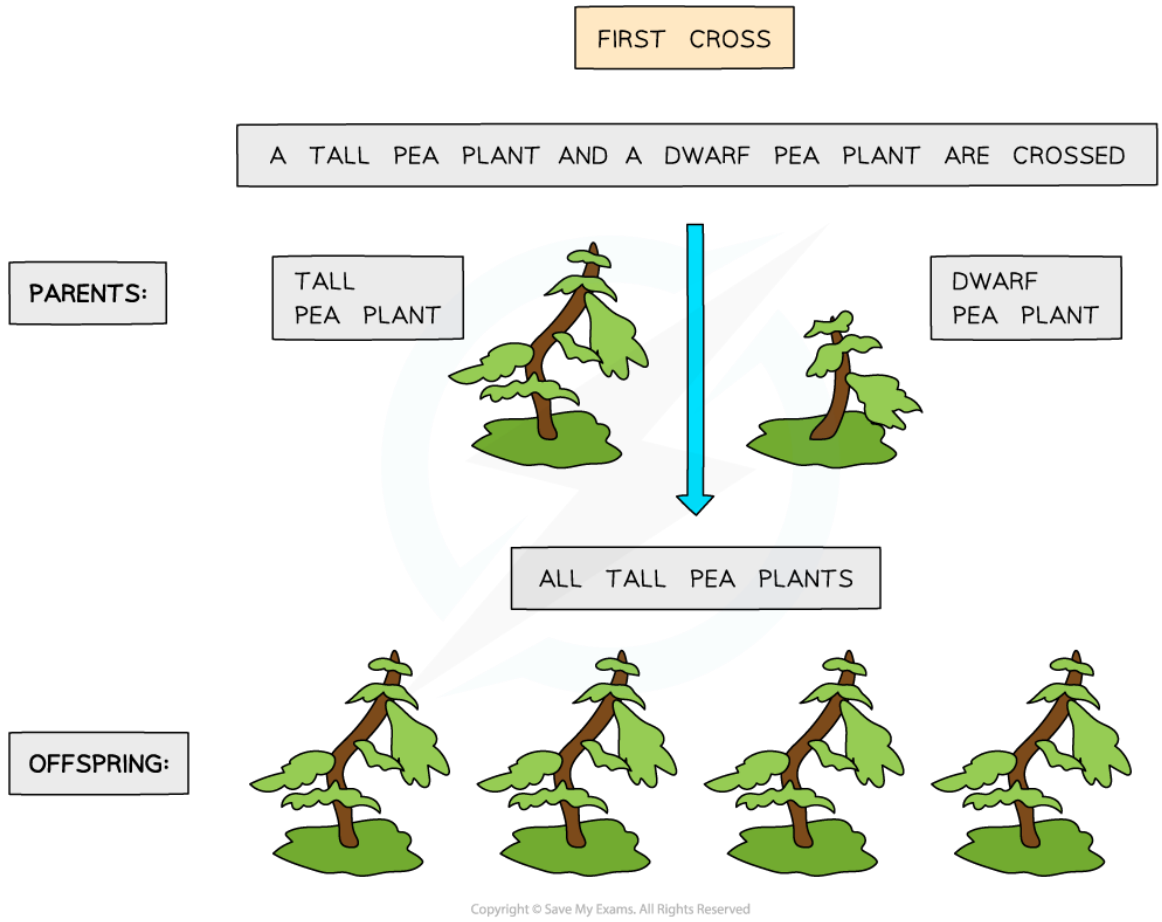
- **Genetics** (遺傳學) is the study of **heredity** (遺傳) and **variations** (變異).
- Heredity is the passing on of characteristics from parents to offspring. Variations are the differences among individuals of a species.

25.1 Mendel's laws of inheritance *(Book 4, p. 25-2)*

A Monohybrid inheritance *(Book 4, p. 25-2)*

1 Mendel's experiments

- Mendel started off with experiments on the inheritance of one pair of **contrasting characters** tall and short stems of garden pea plants. This type of inheritance involving one pair of contrasting characters is called (1) _____ _____ (單基因遺傳).
- Procedure of his experiment:



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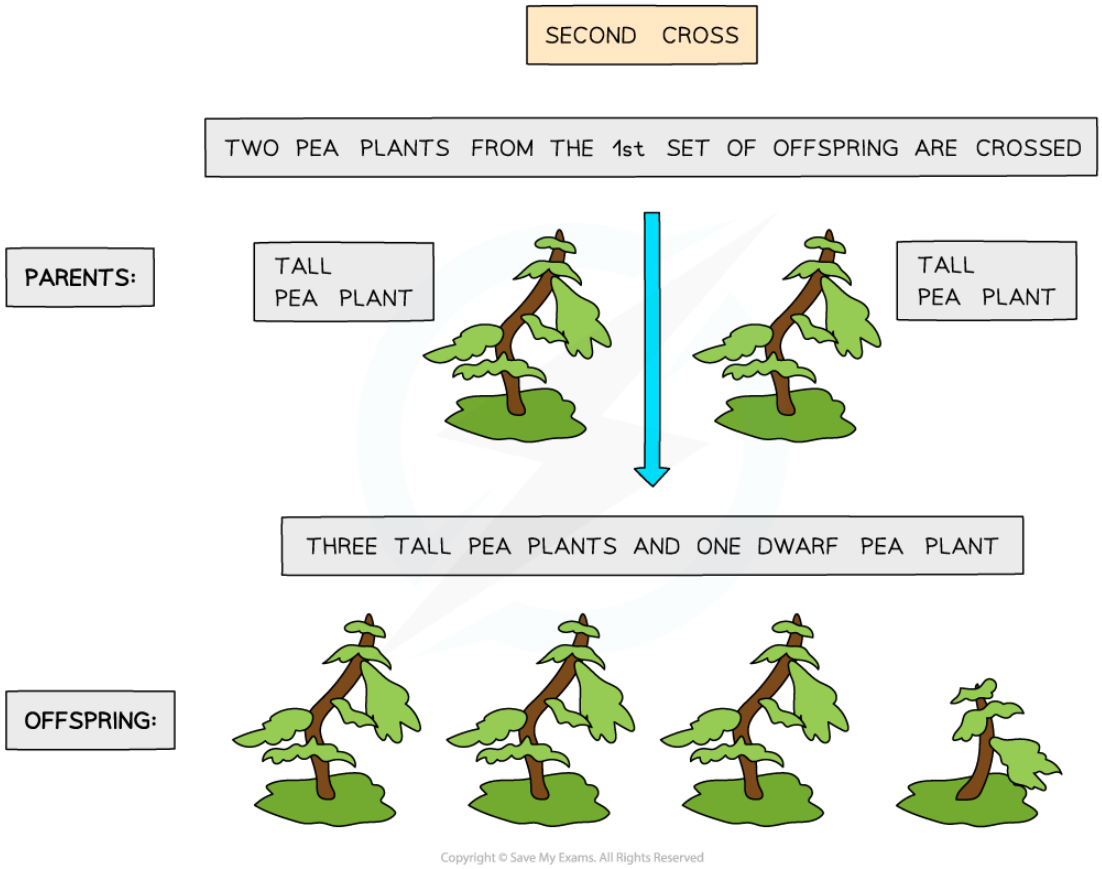
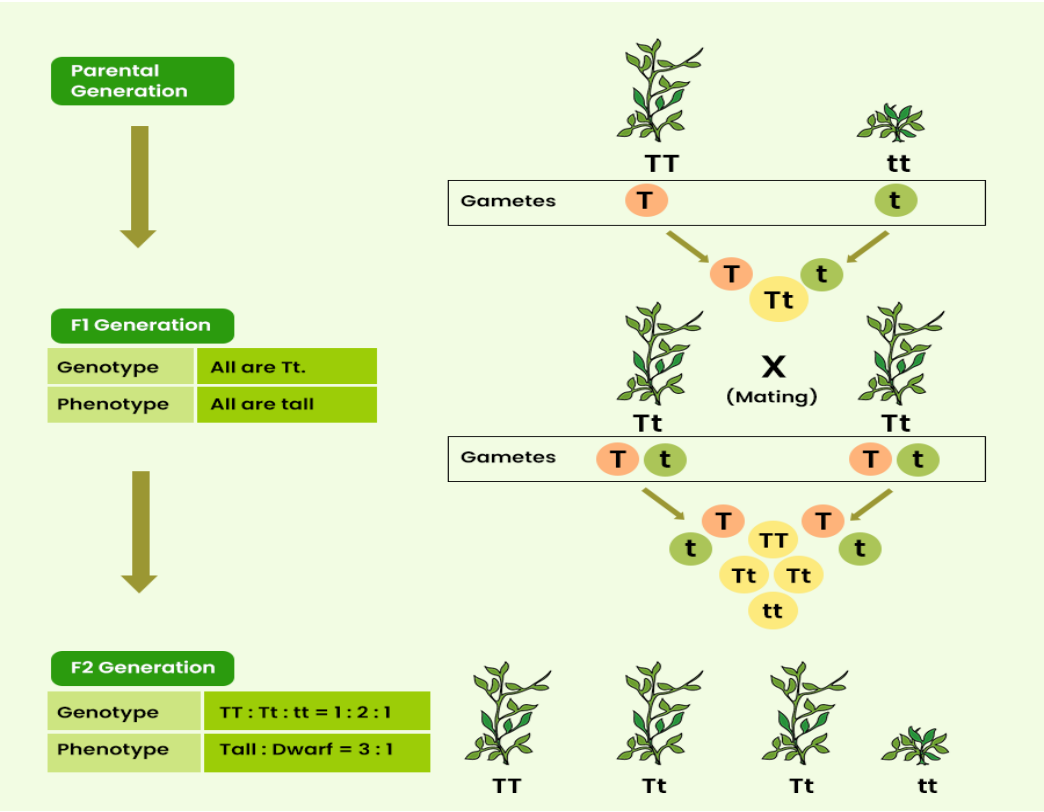


- Summary of the two crosses:



Ratio (2) _____ (3) _____

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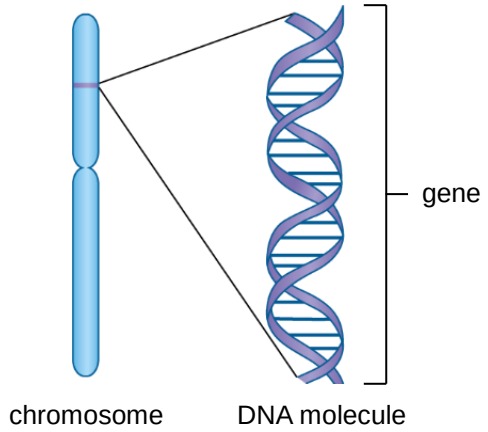
- Mendel also performed other **monohybrid crosses** (單基因雜種雜交) in pea plants. The results of the crosses showed a similar pattern of inheritance:
  - (4) _____ (All / Only one) of the two contrasting characters was seen in the F_1 generation.
  - The character which had disappeared in the F_1 generation (5) _____ in the F_2 generation.
  - The ratio of the two characters in F_2 plants was always about (6) _____.

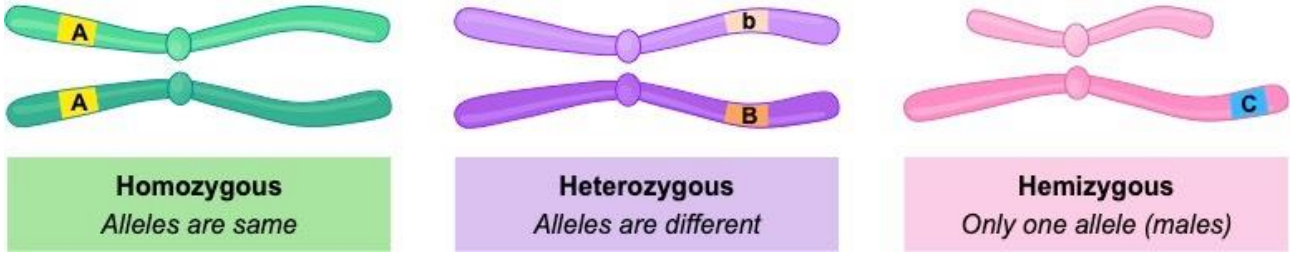
Interpretation of Mendel's experimental results

- Mendel proposed that inherited characteristics are controlled by a pair of 'hereditary factors'.
- Now we know that the **basic unit of heredity** is a (7) _____. A gene is a **segment of a DNA molecule** on a chromosome that controls a particular characteristic.



A gene on a chromosome

- A gene can exist in two or more alternative forms called (8) _____ (等位基因), which occur at the same position on a pair of homologous chromosomes. The position of a gene on a chromosome is called a (9) _____ _____ (基因位點).
  - If the **two alleles are the same**, the individual is said to be (10) _____ (純合的) for the characteristic, and the individual is called a homozygote (純合子).
  - If the **two alleles are different**, the individual is said to be (11) _____ (雜合的) for the characteristic, and the individual is called a heterozygote (雜合子).



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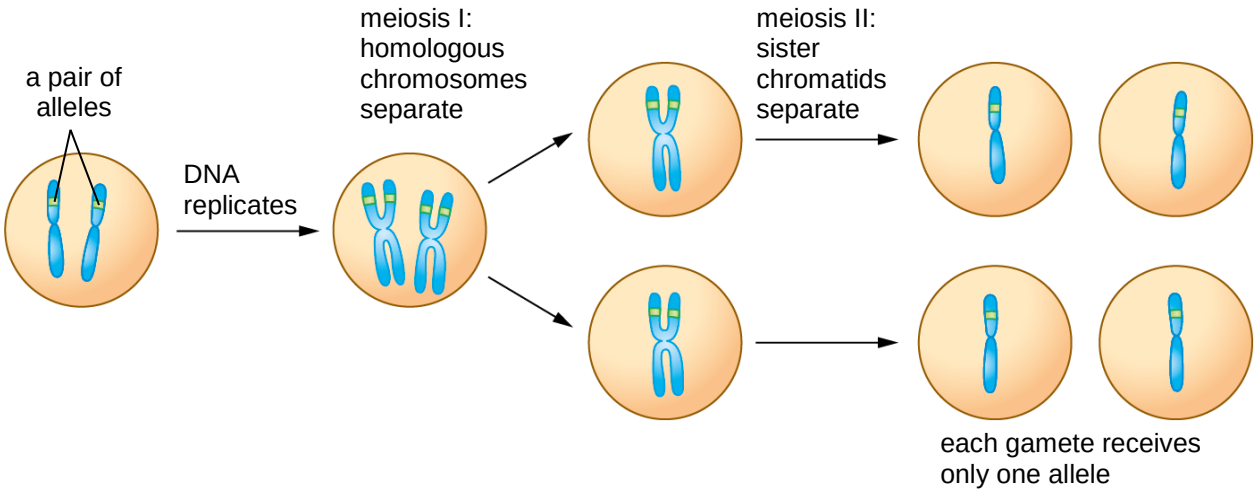
- The results of Mendel's experiment can be explained in terms of alleles, genes, chromosomes and (12) _____ (mitotic / meiotic) cell division.

Why are all F_1 plants tall?

❶ The height of a pea plant is controlled by a pair of (13) _____. We use T and t to represent the alleles for tallness and shortness respectively. Since the parent plants are pure-bred, the tall plant possesses two tall alleles while the short plant possesses two short alleles. The (14) _____ (基因型) (**genetic make-up**) of them are (15) _____ and (16) _____ respectively.

❷ During gamete formation (i.e. **meiotic cell division**), **alleles** (17) _____ so that **each gamete receives only** (18) _____ **allele** from each pair. This is known as the **Mendel's first law of inheritance**, or the **Law of** (19) _____ (分離定律).



Separation of alleles during gamete formation (only one pair of alleles is shown)

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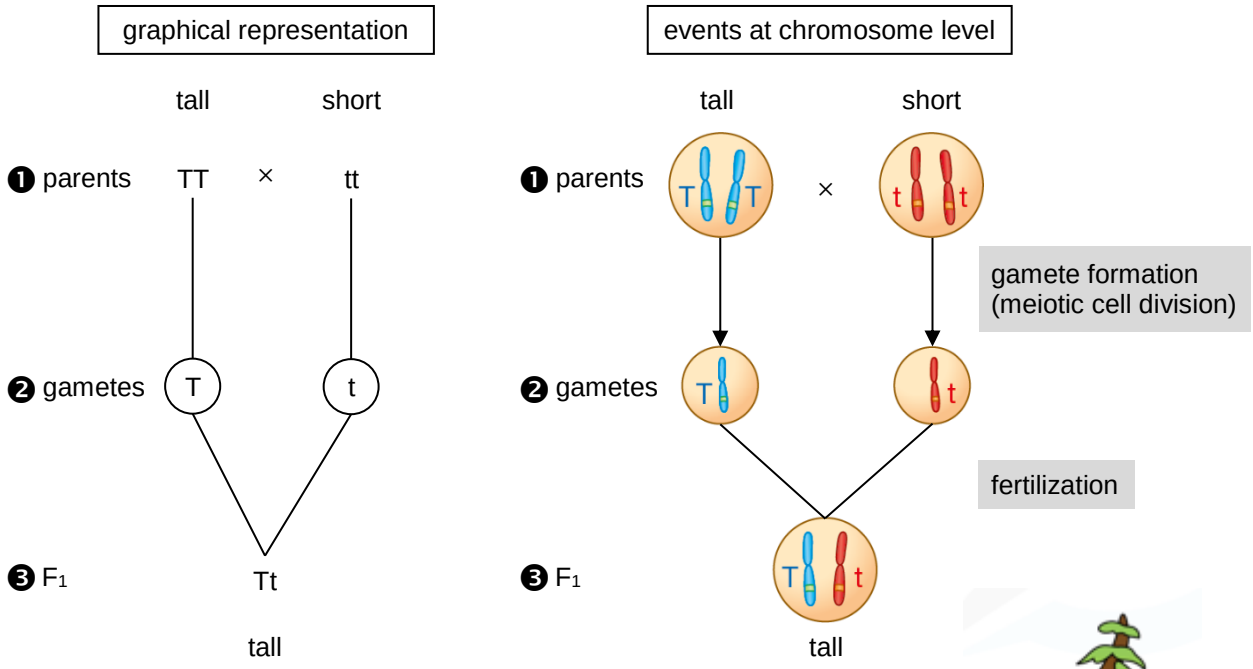
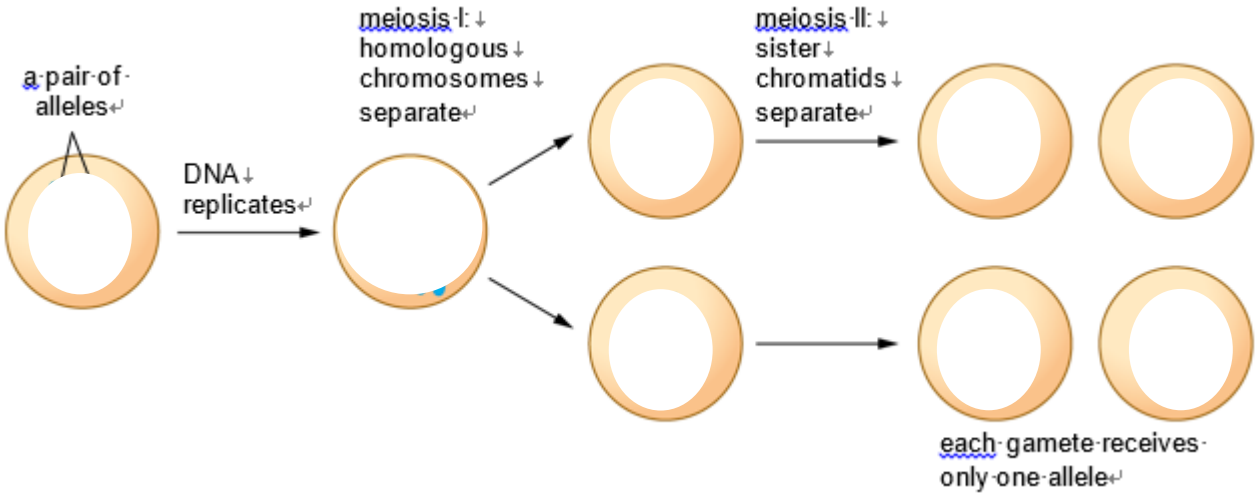
❸ Fertilization of gametes from the two parent plants results in the F_1 generation with the allele combination (20) _____. The (21) _____ (表現型) (**observable character**) of these plants is tall because allele T masks the expression of allele t. Allele T is said to be (22) _____ (顯性) while allele t is (23) _____ (隱性). F_1 plants are also called (24) _____ (雜種).



Explanation of the cross between parent plants in Mendel's experiment

Why do the characters of F_2 plants show a ratio of about 3 : 1?

❹ As the F_1 plants possess two different alleles for the height of a plant, they produce (25) _____ (one / two) **type(s) of gametes** that contain either allele T or t.



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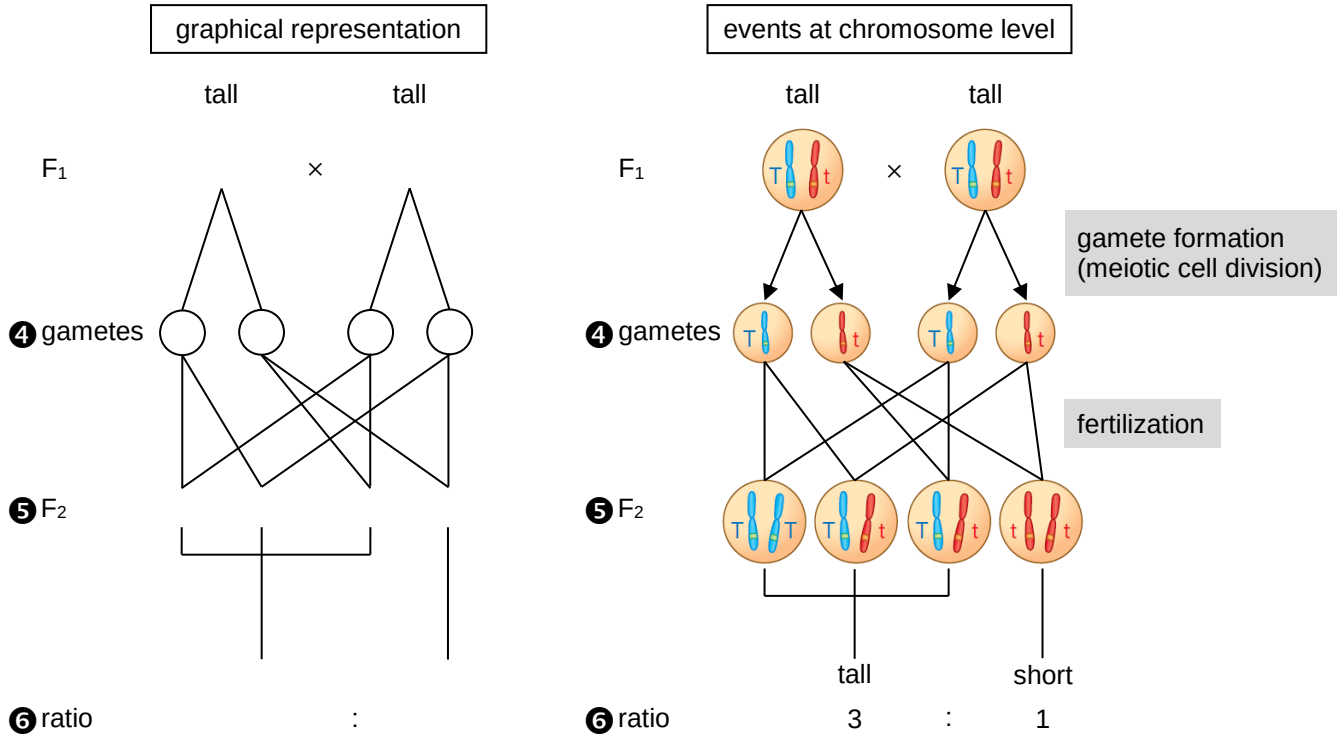
❺ There are (26) _____ **possible combinations** of alleles when **random fertilization** of gametes from two F_1 parents occurs.

❻ A (27) _____ allele can express itself in both homozygous and heterozygous conditions, while a (28) _____ allele can only express itself in a homozygous condition.



▲ Explanation of the cross between F_1 plants in Mendel's experiment

2 Other examples of monohybrid inheritance

	Shape of hairline	Earlobe	Shape of thumb
Dominant character	Widow's peak	Free earlobe	Straight thumb
Recessive character	Straight hairline	Attached earlobe	Hitchhiker's thumb

Practical 25.1 Study inheritance using maize cobs with grains of different colours

(Book 4, p. 25-8; Practical Workbook for SBA 3 • 4, p. 25-1)

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3 Graphical representation of a monohybrid cross

- (29) _____ _____ (遺傳圖解) and (30) _____ _____ (龐氏表) can be used to predict the possible genotypes or phenotypes of the offspring and the probability of each genotype or phenotype appearing in the offspring.

Skill builder

Drawing a genetic diagram *(Book 4, p. 25-10)*

Skill builder

Drawing a Punnett square *(Book 4, p. 25-11)*

- It is important to note that:
  1. The outcome of any cross is (31) _____ (related / not related) to that of any other. This is because each cross is an (32) _____ event.
  2. It is possible to predict the (33) _____ **of offspring** that will have a certain genotype or phenotype, provided that the sample size is large.
  3. Genetic diagrams and Punnett squares only give **expected results**. If a (34) _____ (small / large) **number** of offspring are produced, the **observed results** will be very **close to the expected results**.

4 Test cross for monohybrid inheritance

- A (35) _____ _____ (測交) can be carried out to identify whether an organism with a **dominant character** is homozygous or heterozygous. This can be done by **crossing** the organism of an unknown genotype with a (36) _____ _____ organism.

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- Example of a test cross to identify the genotype of a plant having purple flowers (the dominant character) by crossing it with a plant having white flowers (homozygous recessive):
  - Let F represent the allele for purple flowers and f represent the allele for white flowers.

Case 1: If the genotype of the plant being tested is FF

	plant being tested		homozygous recessive plant
	(37) _____	×	(38) _____
gametes	F		f
F_1 genotype	(39) _____		
phenotype	(40) _____ flowers (all offspring)		

Case 2: If the genotype of the plant being tested is Ff

	plant being tested		homozygous recessive plant
	(41) _____	×	(42) _____
gametes	F f		f
F_1 genotype	(43) _____		(44) _____
phenotype	(45) _____ flowers		(46) _____ flowers
ratio	(47) _____	:	(48) _____

- From the above results, we can see that:
  - If **only** plants having **purple flowers** appear in the offspring, the plant being tested is **probably** (49) _____ (homozygous dominant / heterozygous).
  - If plants having **white flowers** appear in the offspring, the plant being tested is **probably** (50) _____ (homozygous dominant / heterozygous).

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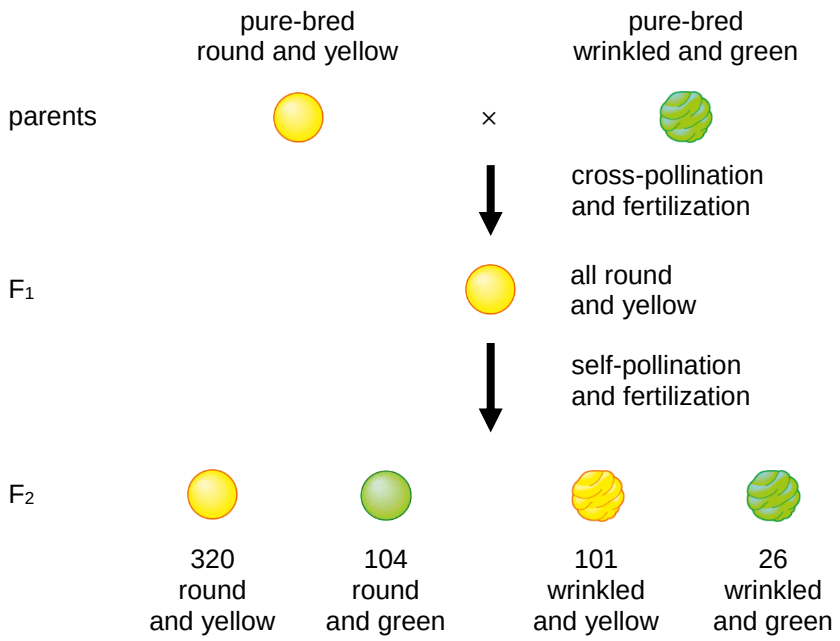
B Dihybrid inheritance *(Book 4, p. 25-15)*

- (51) _____ _____ (雙基因遺傳) is the inheritance of **two pairs of contrasting characters** at the same time.
- Mendel also did experiments on dihybrid inheritance. In one of his experiments, he looked at both the shape and the colour of seed of pea plants:



- Mendel proposed another hypothesis to explain the results. This is known as the **Mendel's second law of inheritance**, or the **Law of** (52) _____ _____ (獨立分配定律). It states that the **separation of alleles** of one gene is **independent** of the separation of alleles of other genes during gamete formation.

Interpretation of Mendel's experimental results

- The shape of seed is controlled by a pair of alleles. The colour of seed is controlled by another pair of alleles **on** (53) _____ (the same / a different) **pair of homologous chromosomes**. We use R to represent the allele for (54) _____ seed and r to represent the allele for (55) _____ seed; Y to represent the allele for (56) _____ seed and y to represent the allele for (57) _____ seed.

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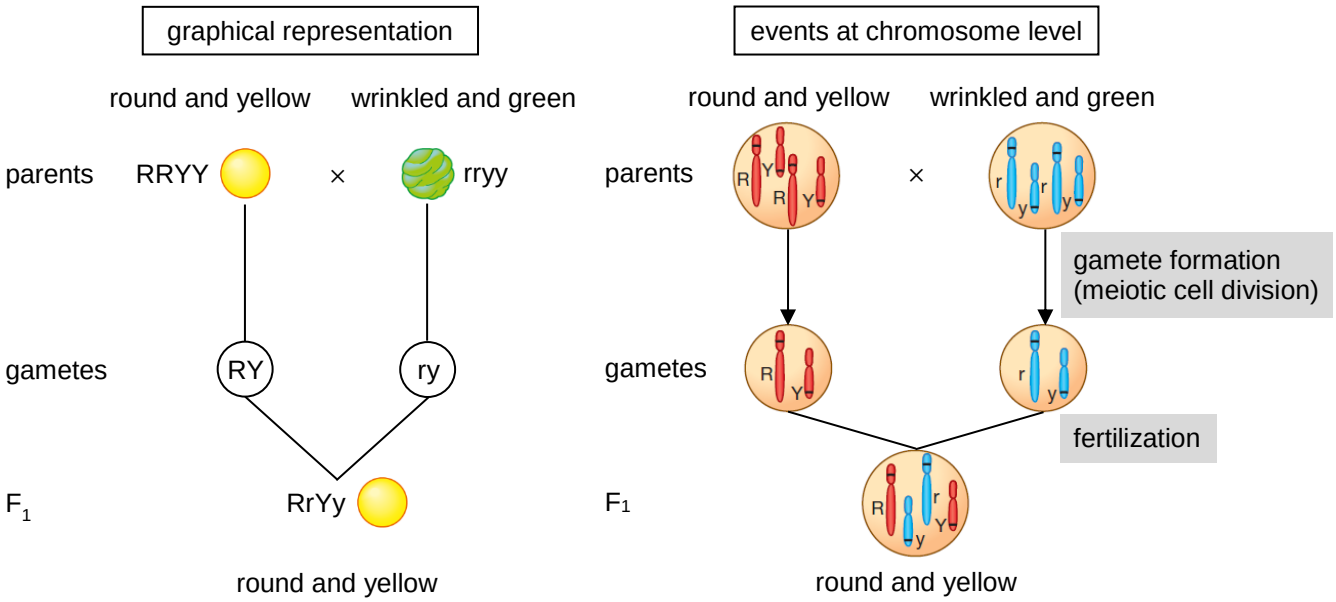
❶ The genotypes of pure-bred parent plants are RRYY and rryy respectively.

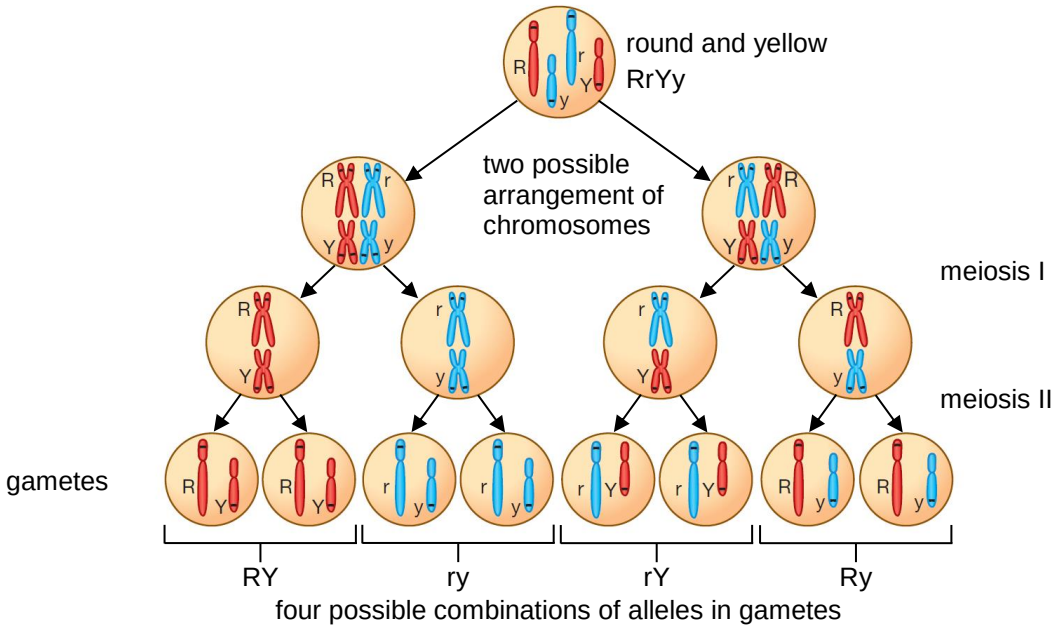
❷ During gamete formation, alleles of each gene separate so that each gamete receives only one allele from each pair. Each parent produces (58) _____ type of gamete (RY or ry).

❸ Fertilization of gametes results in the F_1 generation with the allele combination RrYy. All the F_1 plants have round and yellow seeds because alleles R and Y are dominant over alleles r and y respectively.



▲ Explanation of the cross between parent plants in Mendel's experiment

❹ In the F_1 plants, the alleles of the two genes are **assorted** (59) _____ during gamete formation. This results in the production of (60) _____ types of gametes: RY, Ry, rY and ry.



▲ Alleles of two genes on different homologous chromosomes are assorted independently during gamete formation

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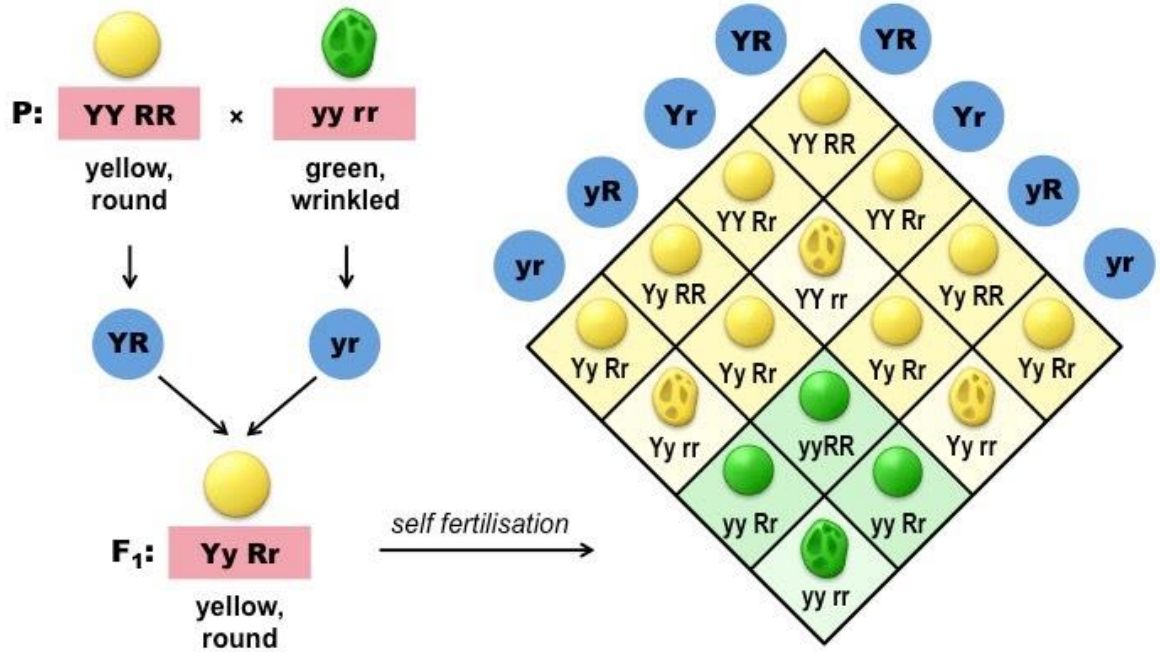
❺ There are (61) _____ possible combinations of alleles when random fertilization of gametes from the two F_1 parents occurs. (62) _____ phenotypes appear in the F_2 generation.

F_2	RY	Ry	♀ rY	ry
RY	RRYY	RRYy	RrYY	RrYy
Ry	RRYy	RRyy	RrYy	Rryy
♂ rY	RrYY	RrYy	rrYY	rrYy
ry	RrYy	Rryy	rrYy	rryy

Phenotypic ratio	round and yellow (R-Y-)	:	round and green (R-yy)	:	wrinkled and yellow (rrY-)	:	wrinkled and green (rryy)
	(63) _____	:	(64) _____	:	(65) _____	:	(66) _____

➤ Punnett square showing the possible combinations of alleles in the F_2 generation



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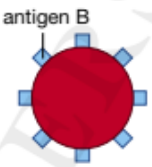
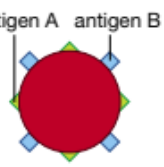
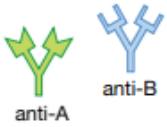
25.2 Inheritance in humans *(Book 4, p. 25-19)*

A Inheritance of blood groups *(Book 4, p. 25-19)*

- Blood groups are determined by the type of (1) _____ (抗原) on the surface of red blood cells.
- **Anti-A antibodies** will **act against antigen A** and **anti-B antibodies** will **act against antigen B**, causing (13) _____ (聚集) of blood cells or (14) _____ (溶血).
- If **recipient's blood contains antibodies** that will **act against the antigens** on the red blood cells of the donor's blood, the donor's blood is said to be (15) _____ (不相容) with the recipient's blood.

Blood group	A	B	AB	O
Antigen on red blood cells				
Antibody in plasma			none	
Can receive blood from people with	Blood groups A and O	Blood groups B and O	Blood groups A, B, AB and O	Blood group O
Can donate blood to people with	Blood groups A and AB	Blood groups B and AB	Blood group AB	Blood groups A, B, AB and O

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- The production of antigens A and B is controlled by a single gene. The gene has three (16) _____: **I^A**, **I^B** and **i**. Genes with more than two alternative forms are said to have (17) _____ _____ (多等位基因).

Allele	Responsible for the production of…	Dominant or recessive
I^A	Antigen A	(18) _____-_____ (等顯性)
I^B	Antigen B	
i	Not responsible for producing antigens	Recessive

- Possible genotypes of people of different blood groups:

Blood group (phenotype)	**A**	**B**	**AB**	**O**
Antigen produced	Antigen (19) ___	Antigen (20) ___	Antigens A and B	(21) _____
Possible genotype	I^AI^A and I^Ai	I^BI^B and I^Bi	I^AI^B	ii

- The possible blood groups of the baby of a man who is **heterozygous** for blood group A and his wife who is heterozygous for blood group B:

	blood group A		blood group B	
Parents	(22) _____	×	(23) _____	
Gametes	I^A i		I^B i	
F_1 genotype	(24) _____	(25) _____	(26) _____	(27) _____
Phenotype	blood group (28) _____	blood group (29) _____	blood group (30) _____	blood group (31) _____
Ratio	(32) _____ :	(33) _____ :	(34) _____ :	(35) _____

- From the genetic diagram, we can see that the baby will have a 25% chance of being blood groups A, B, AB and O respectively.

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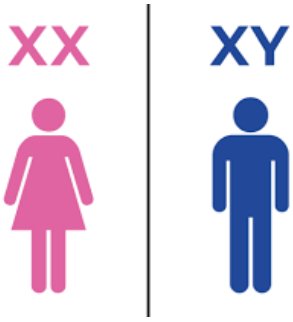
B Sex determination *(Book 4, p. 25-22)*



- In the somatic cells in our body, there are 23 pairs of homologous chromosomes. Of the 23 pairs, **22 pairs** are the same in males and females. They are called (36) _____ (常染色體). The two members of each pair of autosomes have **same size and same shape**.
- The **23rd pair** of chromosomes are called the (37) _____ **chromosomes** (性染色體). They determine the sex of an individual.
  - Each somatic cell of a **male** contains **one X chromosome** and **one Y chromosome**.
  - Each somatic cell of a **female** contains **two** (38) _____ (X / Y) **chromosomes**.
- The **sex of a baby** is **determined by the type of** (41) _____ that is involved in fertilization:
  - If the ovum is fertilized by an **X-bearing sperm**, the zygote develops into a (42) _____.
  - If the ovum is fertilized by a **Y-bearing sperm**, the zygote develops into a (43) _____.

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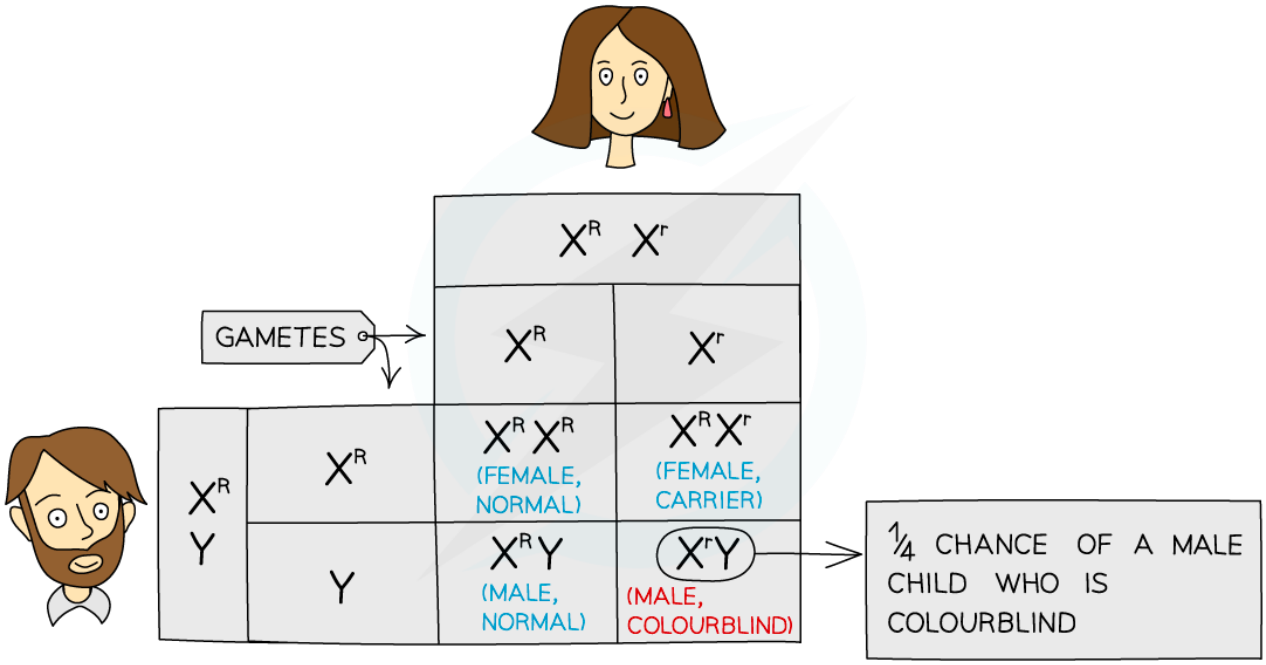
C Inheritance of sex-linked characteristics *(Book 4, p. 25-24)*

- Genes on the sex chromosomes and the characteristics they control are said to be (46) _____-_____ (性連鎖). The sex-linked genes that are located on the X chromosome are called (47) _____-_____ **genes**.
- **Red-green colour blindness** is a genetic disorder caused by a recessive allele on the X chromosome.

	Genotype	Phenotype
Female	X^RX^R	Normal colour vision
	X^RX^r	Normal colour vision (she is a carrier 帶基因者 of the allele for red-green colour blindness)
	(48) _____	Red-green colour-blind
Male	(49) _____	Normal colour vision
	(50) _____	Red-green colour-blind

(Let R represent the allele for normal colour vision and b represent the allele for red-green colour blindness.)



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- Red-green colour blindness is more common in (51) _____ (males / females).
- Genetic diagrams can be used to show how alleles for red-green colour blindness in male and female parents are passed on to the offspring.

Case 1: Red-green colour-blind father × mother with normal colour vision

	red-green colour-blind male		normal colour vision female
Parents	(52) _____	×	(53) _____
Gametes	X^r Y		X^R
F_1 genotype	(54) _____		(55) _____
Phenotype	(56) _____ (female carrier)		(57) _____ (male)
Ratio	(58) _____	:	(59) _____

- A father must pass his (60) _____ (X / Y) chromosome on to his daughters. The **daughters must receive an allele** for the disease from a diseased father.
- A father must pass his (61) _____ (X / Y) chromosome on to his sons. **The sons** (62) _____ (will / will not) **receive an allele for the disease** from a diseased father.

Case 2: Father with normal colour vision × red-green colour-blind mother

	normal colour vision male		red-green colour-blind female
Parents	(63) _____	×	(64) _____
Gametes	X^R Y		X^r
F_1 genotype	(65) _____		(66) _____
Phenotype	(67) _____ (female carrier)		(68) _____ (male)
Ratio	(69) _____	:	(70) _____

- **Both the sons and daughters receive an allele** for the disease from a diseased mother.
- The sons' X chromosomes must come from their (71) _____. **All the sons of a diseased mother must have the disease**.
- **Daughters will have normal colour vision if their father has normal colour vision**.

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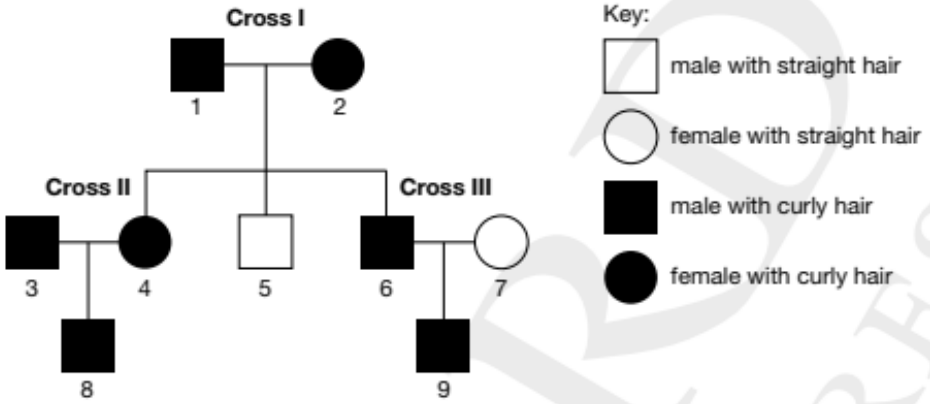
D Studying inheritance in humans using pedigrees *(Book 4, p. 25-27)*

- We can obtain information about inheritance in humans by studying (72) _____ (譜系).

The pedigree below shows the inheritance of curly and straight hair in a family. The characteristic is controlled by a pair of alleles on autosomes.



a **i** Which of the crosses allows you to deduce which character is dominant? (1 mark)

ii Without using genetic diagrams, deduce which character is dominant. (5 marks)

b Without using genetic diagrams, deduce the possible genotype of individual 9. (4 marks)

- By tracing the pattern of inheritance of a characteristic through a pedigree, we can:
  - find out if the allele for a character is dominant or recessive.
  - deduce the possible genotypes of a certain individual.

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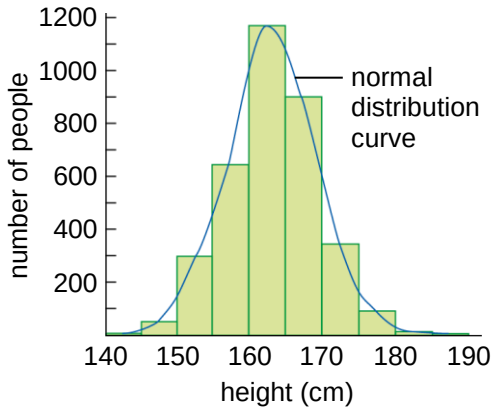
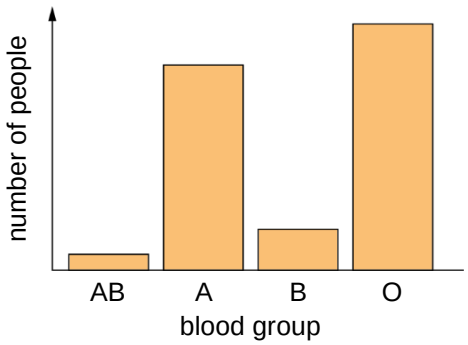
25.3 Variations in organisms *(Book 4, p. 25-34)*

- The differences among individuals of the same species are called (1) _____.

A Types of variations *(Book 4, p. 25-34)*

- There are two types of variations: (2) _____ _____ (連續變異) and (3) _____ _____ (不連續變異):

	Continuous variation	Discontinuous variation
Number of phenotypes	A **continuous range of** (4) _____ **phenotypes** between two extremes	A few (5) _____ phenotypes with **no intermediates**
Cause	Genetic and environmental factors	Mainly genetic factors
Genetic control	Usually controlled by (6) _____ (one or a few / many) **gene(s)**	Usually controlled by (7) _____ (one or a few / many) **gene(s)**
Representative graph	(8) _____ (直方圖), from which a (9) _____ _____ **curve** can be obtained 	(10) _____ **chart** 
Examples in humans	Height, weight, skin colour, hand span, foot length, heart rate, blood pressure and intelligence quotient (IQ)	Shape of hairline, free earlobes or attached earlobes, ability to roll the tongue, straight thumb or hitchhiker's thumb, having or not having dimples and ABO blood groups

Practical 25.2 Observation and analysis of variations in humans
(Book 4, p. 25-36; Practical Workbook for SBA 3 • 4, p. 25-3)

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B Causes of variations *(Book 4, p. 25-37)*

- Variations in organisms are caused by (11) _____ and/or (12) _____ **factors**.

1 Genetic factors

- The genetic differences between individuals are called (13) _____ _____ (遺傳變異), which are caused by the following four processes:
  - i) (14) _____ _____ of chromosomes at meiosis
  - ii) (15) _____ _____ at meiosis
  - iii) Random combination of gametes during (16) _____
  - iv) (17) _____ (突變), which refers to the **sudden and permanent change in the DNA** of an organism. **Only mutations** that occur **in gametes or gamete-producing cells can be passed on to the next generation**.
- Of the four processes, only mutations may produce new (18) _____. They are the ultimate source of genetic variation. The other three processes only lead to different (19) _____ of the existing alleles.

2 Environmental factors

- All organisms are affected by the environment in which they live. For example, the colour of garlic chives (韭菜) is affected by the presence of light. When they grow in darkness, they become yellow in colour, and are known as yellow chives (韭黃).

Garlic chives and yellow chives

- Characteristics showing **continuous variation** are (20) _____ (less / more) **likely** to be affected by the environmental factors, while those showing discontinuous variation are (21) _____ (less / more) likely to be affected.

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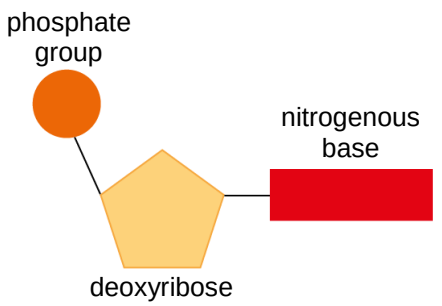
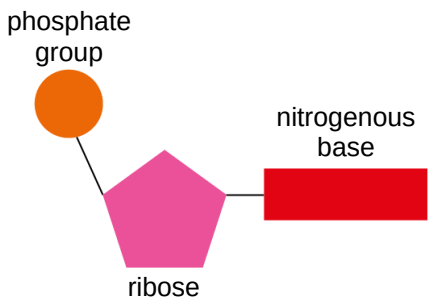
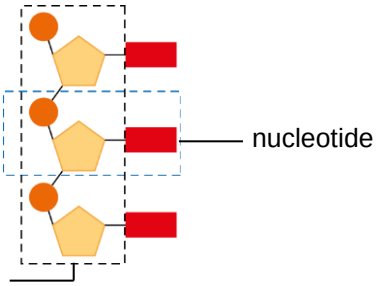
25.4 DNA – the carrier of genetic information *(Book 4, p. 25-41)*

1 Chemical structure of nucleic acids

- **Nucleic acids** (核酸) include:
  - **DNA** (**d**eoxyribo**n**ucleic **a**cid),
  - **RNA** (**r**ibo**n**ucleic **a**cid) (核糖核酸), which is important in (1) _____ synthesis.
- A nucleic acid is a long chain of (2) _____ (核苷酸). Each nucleotide consists of a **phosphate group**, a **5-carbon sugar** and a (3) _____ _____ (含氮鹼基). The nucleotides composing DNA and RNA are different.

	Nucleotide of DNA (deoxyribonucleotide 脫氧核糖核苷酸)	**Nucleotide of RNA (ribonucleotide 核糖核苷酸)**
		
Type of sugar	▪ (4) _____ (脫氧核糖)	▪ (5) _____ (核糖)
Type of bases	▪ (6) _____ (**A**) (腺嘌呤) ▪ (7) _____ (**T**) (胸腺嘧啶) ▪ (8) _____ (**C**) (胞嘧啶) ▪ (9) _____ (**G**) (鳥嘌呤)	▪ (10) _____ (A) ▪ (11) _____ (**U**) (尿嘧啶) ▪ (12) _____ (C) ▪ (13) _____(G)

- The nucleotides join to form a (14) _____ (多核苷酸):



(15) _____-_____
backbone

- An (16) _____ (DNA / RNA) **molecule** consists of a **single strand** of polynucleotide chain, while a (17) _____ (DNA / RNA) **molecule** consists of **two strands** of polynucleotide chains.

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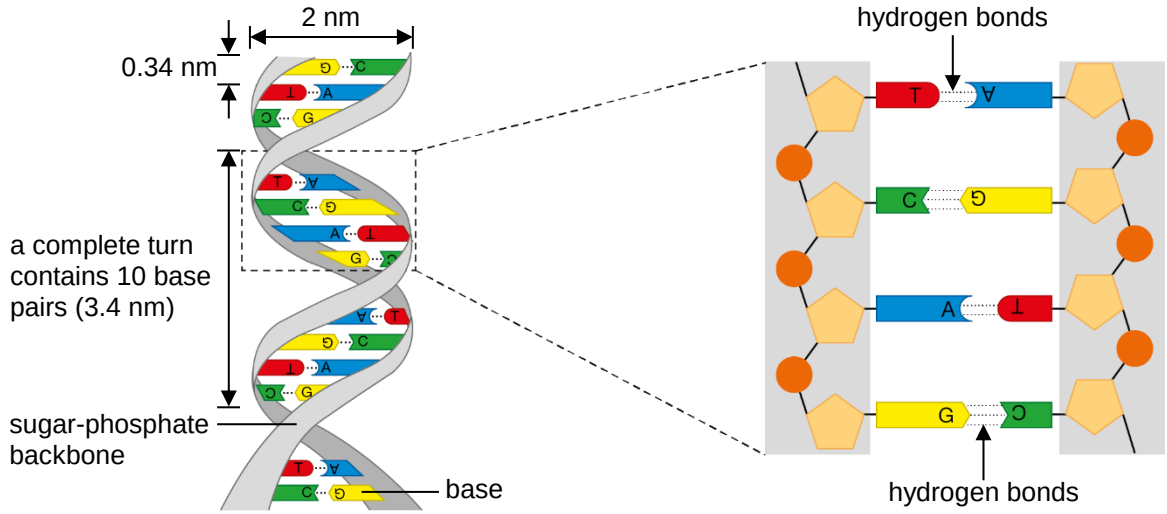
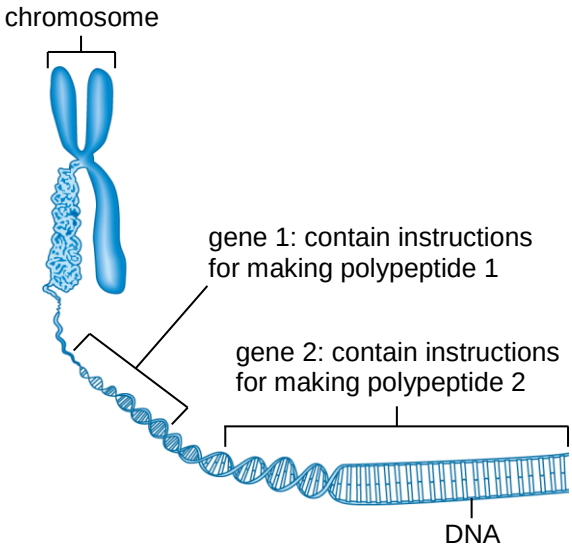
- In 1953, James Watson and Francis Crick proposed that in a DNA molecule:
  - two polynucleotide chains run in (18) _____ **directions** and twist around each other to form a (19) _____ _____.
  - **A** of one chain **pairs** only **with** (20) _____ of other chain by (21) _____ **bonds**, and **C with** (22) _____. These rules are called (23) _____ _____ _____ (互補鹼基配對).



Structure of DNA

2 How DNA determines body characteristics

- A (24) _____ is a segment of a DNA molecule on a chromosome.
- The **base sequence** of DNA in a gene determines the (25) _____ _____ **sequence** in a polypeptide in a way called the (26) _____ _____ (遺傳密碼).



A gene is a segment of a DNA molecule on a chromosome

- Genes can determine the **body characteristics** of an organism by controlling the **types of** (27) _____ a cell makes. Each inherited characteristic is controlled by a gene or several genes.

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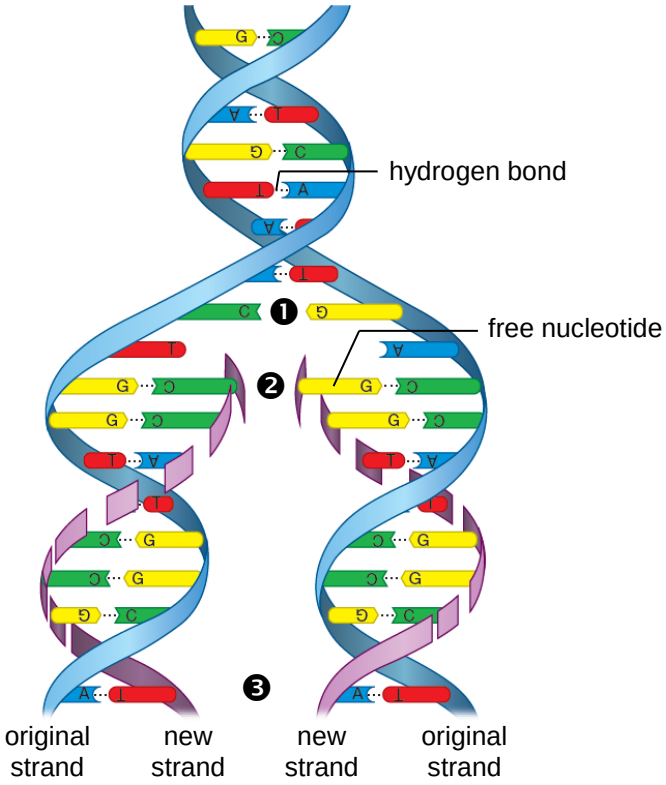
3 How the structure of DNA enables it to function

- DNA is a **large molecule** consisting of a large number of (28) _____. This allows DNA to carry a **large amount of** (29) _____ **information**.
- DNA is a (30) _____ **molecule**. Its helical structure is maintained by many hydrogen bonds between the two strands.
- The base pairs project inward in a DNA molecule. This **protects the genetic information** from (31) _____.
- The pairing of specific bases enables **accurate** (32) _____ (複製) of DNA so that the same genetic information can be passed from generation to generation.

The process of DNA replication



❶ The hydrogen bonds between two DNA strands (33) _____. The two strands separate.

❷ Each DNA strand acts as a (34) _____ (模板) for the synthesis of a new DNA strand. The bases in free nucleotides form complementary base pairing with the bases in the DNA strands. **A pairs with T**, **C pairs with G**. An enzyme called (35) _____ _____ (DNA 聚合酶) catalyses the joining of the adjacent nucleotides to form a new DNA strand.

❸ Two (36) _____ (identical / different) DNA molecules are formed. Each contains an original strand and a new strand.

Practical 25.3 Extraction of DNA

(Book 4, p. 25-46 Practical Workbook for SBA 3 • 4, p. 25-10)

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Answers

Ch 25 Basic genetics

25.1

1	monohybrid inheritance			**2**	3	**3**	1
4	Only one	**5**	reappeared	**6**	3 : 1	**7**	gene
8	alleles	**9**	gene locus	**10**	homozygous	**11**	heterozygous
12	meiotic	**13**	alleles	**14**	genotypes	**15**	TT
16	tt	**17**	separate	**18**	one	**19**	Segregation
20	Tt	**21**	phenotype	**22**	dominant	**23**	recessive
24	hybrid	**25**	two	**26**	four	**27**	dominant
28	recessive	**29**	Genetic diagrams	**30**	Punnett squares	**31**	not related
32	independent	**33**	proportions	**34**	large	**35**	test cross
36	homozygous recessive			**37**	FF	**38**	ff
39	Ff	**40**	purple	**41**	Ff	**42**	ff
43	Ff	**44**	ff	**45**	purple	**46**	white
47	1	**48**	1	**49**	homozygous dominant		
50	heterozygous	**51**	Dihybrid inheritance	**52**	Independent Assortment		
53	a different	**54**	round	**55**	wrinkled	**56**	yellow
57	green	**58**	one	**59**	independently	**60**	four
61	16	**62**	Four	**63**	9	**64**	3
65	3	**66**	1				

25.2

1	antigens	**2**	A	**3**	B	**4**	None
5	B	**6**	A	**7**	None	**8**	O
9	O	**10**	A	**11**	AB	**12**	AB
13	clumping	**14**	haemolysis	**15**	incompatible	**16**	alleles
17	multiple alleles	**18**	Co-dominant	**19**	A	**20**	B
21	None	**22**	I^Ai	**23**	I^Bi	**24**	I^AI^B
25	I^Ai	**26**	I^Bi	**27**	ii	**28**	AB
29	A	**30**	B	**31**	O	**32**	1
33	1	**34**	1	**35**	1	**36**	autosomes

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37	sex	**38**	X	**39**	only one	**40**	two
41	sperm	**42**	girl	**43**	boy	**44**	random
45	the same	**46**	sex-linked	**47**	X-linked	**48**	X^bX^b
49	X^BY	**50**	X^bY	**51**	males	**52**	X^bY
53	X^BX^B	**54**	X^BX^b	**55**	X^BY	**56**	normal colour vision
57	normal colour vision	**58**	1	**59**	1	**60**	X
61	Y	**62**	will not	**63**	X^BY	**64**	X^bX^b
65	X^BX^b	**66**	X^bY	**67**	normal colour vision		
68	red-green colour-blind			**69**	1	**70**	1
71	mother	**72**	pedigrees	**73**	generation	**74**	Males
75	females	**76**	marriage				

25.3

1	variations	**2**	continuous variation	**3**	discontinuous variation		
4	intermediate	**5**	distinct	**6**	many	**7**	one or a few
8	Histogram	**9**	normal distribution	**10**	Bar		
11	genetic / environmental			**12**	environmental / genetic		
13	genetic variations	**14**	Independent assortment			**15**	Crossing over
16	fertilization	**17**	Mutation	**18**	alleles	**19**	combinations
20	more	**21**	less				

25.4

1	protein	**2**	nucleotides	**3**	nitrogenous base	**4**	Deoxyribose
5	Ribose	**6**	Adenine	**7**	Thymine	**8**	Cytosine
9	Guanine	**10**	Adenine	**11**	Uracil	**12**	Cytosine
13	Guanine	**14**	polynucleotide	**15**	sugar-phosphate	**16**	RNA
17	DNA	**18**	opposite	**19**	double helix	**20**	T
21	hydrogen	**22**	G	**23**	complementary base pairing		
24	gene	**25**	amino acid	**26**	genetic code	**27**	proteins
28	nucleotides	**29**	genetic	**30**	stable	**31**	damage
32	replication	**33**	break	**34**	template	**35**	DNA polymerase
36	identical						

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